

RECONPRO

FULL CODEBASE INTELLIGENCE REPORT

20-Part Exhaustive Intelligence Analysis
200+ Files | 121,000+ Lines of Code | 28 Modules | 50+ CLI Commands

Classification	Confidential	Version	v11.0.0
Date	2026-08-11	Python	3.10+ Pure Stdlib
Methodology	Full File Inspection	LOC Analyzed	121,000+

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PART 1: Executive Summary

ReconPro is a pure-Python enterprise security reconnaissance framework comprising 121,000+ lines of code across 200+ source files. The system operates with zero external runtime dependencies (Python stdlib only), requiring only Python 3.10+ and optionally the Rich library for terminal rendering. The codebase implements 28 scanning modules, 3 intelligence engines, 50+ CLI commands, 6 enterprise integrations, and a full Textual-based TUI application. Three dual output channels serve operators: a synchronous CLI scanner, an asynchronous concurrent engine, and a REST API server with token-based authentication.

The architecture follows a layered design with 7 levels (L0 Constants through L6 Entry Points), centered on a universal Finding dataclass that standardizes output across all modules. A centralized registry pattern in registry.py eliminates circular dependencies by providing a single source of truth for 24 remote and 3 local modules. The three intelligence systems (AI Analyst, Attack Graph, Threat Intel) compose into a unified Intelligence Pipeline that produces composite risk scores, attack paths, and threat correlations post-scan.

Performance benchmarks reveal that the Attack Graph engine is the primary bottleneck at scale, exhibiting $O(n^2)$ growth. At 2000 findings, the full intelligence pipeline takes approximately 37 seconds, with the Attack Graph consuming 87.6% of total execution time. The AI Analyst scales at approximately $O(n^{1.3})$ and Threat Intel maintains linear $O(n)$ scaling. Memory usage remains manageable even at scale, peaking at approximately 24MB for 2000 findings.

Metric	Value	Metric	Value
Total Files	200+	Total LOC	121,000+
Python Modules	28	CLI Commands	50+
Intelligence Engines	3	Integrations	6
Test Files	29	Test Methods	1,250+
REST API Endpoints	20+	TUI Widgets	7
Report Formats	5	Scoring Systems	3

Production Readiness Assessment

A prior 10-team engineering review assigned ReconPro a Production Readiness Score of 91/100 (Grade A). The architecture supports Docker containerization with production-grade Kubernetes manifests including security contexts, resource limits, health probes, topology-aware routing, and pod anti-affinity. Six security fixes have been applied addressing path traversal, XSS injection, severity sort inversion, shared mutable state, and silent exception swallowing. However, critical issues remain including version string inconsistency across 7+ files, dual scan engine divergence, and 274 lines of duplicated constant definitions.

PART 2: Architecture Overview

ReconPro employs a 7-layer architecture (L0 through L6) that enforces strict dependency directionality. L0 (Constants) provides shared definitions including severity levels, grade thresholds, color mappings, and path constants. L1 (Utilities) offers 17 pure functions for target validation, severity normalization, score computation, and cryptographic operations. L2 (Core) contains the universal Finding dataclass, RateLimiter, and HTTP probe abstraction that every module depends on. L3 (Engine) houses the dual scan engines (synchronous scanner.py and asynchronous engine.py) along with the module registry. L4 (Intelligence) implements the three analysis engines and their orchestration pipeline. L5 (Outputs) handles all report generation (PDF, HTML, JSON, Markdown, SARIF, XML, CSV). L6 (Entry Points) provides CLI, TUI, REST API, chat mode, and agent interfaces.

Key Architectural Decisions

ADR	Decision	Status	Files Affected
ADR-001	Pure Python stdlib only	Accepted	All modules
ADR-002	Centralized module registry	Accepted	registry.py, scanner.py
ADR-003	HTTP timing OS fingerprinting	Accepted	quantum_fingerprint.py
ADR-004	Universal Finding dataclass	Accepted	http_layer.py, all modules
ADR-005	Zero-config operation	Accepted	constants.py, cli.py

Design Patterns

The codebase implements several well-established design patterns. The Registry Pattern centralizes module definitions in a single file, eliminating circular imports between scanner.py and individual modules. The Strategy Pattern governs proxy rotation with four strategies (Round Robin, Random, Least Connections, Geo-Distributed). The Observer Pattern appears in the event-driven scan engine, where ScanEvent objects are emitted to EventCollector callbacks. The Template Method Pattern structures defense generation with category-specific fix templates. Lazy imports are used extensively throughout, with over 30 modules employing deferred import inside function bodies to prevent circular dependencies and reduce startup time. Duck typing is pervasive, with utility functions accepting both Finding objects and plain dictionaries via hasattr/getattr patterns.

Dependency Flow

The dependency graph is strictly hierarchical with one universal dependency: every scanning module imports from http_layer.py (L2) for the Finding dataclass and http_probe function. Cross-module dependencies are minimal and always lazy-loaded via try/except blocks. Five modules import sibling packages: bot.py imports threat_feeds, geoip, tunnel_detect, and exfil_channels; chain.py imports cross_validator and drift_monitor; recon.py imports wishes, social_graph, and sovereignty; oblivion.py imports zai_stream, wishes, ai_red_team, attribution, and cognitive_sec; and gorgon.py imports zai_stream, wishes, ai_red_team, ai_cve_db, kill_chain, and shadow_it. All cross-module imports are wrapped in try/except, ensuring graceful degradation when sibling packages are unavailable.

PART 3: Module Registry and Organization

The module registry system in `registry.py` serves as the single source of truth for all scanning modules. It defines 24 remote scanning modules, 3 local audit modules, and maintains default module lists for standard operations. The lazy-loaded `_get_runners()` function caches all module runner functions on first access, preventing repeated imports. However, since `MODULE_REGISTRY` is assigned at module level, all 25 modules are actually imported when `registry.py` is first imported, effectively defeating the lazy-loading pattern.

Category	Count	Modules	Default
Core Remote	11	recon, auth, chain, bot, gorgon, oblivion, vibesec, nhi, pegasus, cloud_recon, team	Core Remote
Advanced Remote	12	quantum_fingerprint, dark_web_monitor, info_ops, steg, covert_channel, zeroday_hunter, infra_ghost, sigint, attributor, w	Advanced Remote
Local Audit	3	host, dev, doctor	All 3
Special	2	iac_audit (Terraform/K8s/Docker), container_sec (escape analysis)	Excluded

Adding a new module requires changes to three files: the module file itself in `modules/`, the `__init__.py` re-export list, and the `registry.py` `build_module_registry` function. Each module must expose a `run_*(target, base_url, timeout, verify_tls) -> List[Finding]` entry point. The `vibesec` module is uniquely handled across the codebase because it returns a 4-tuple (`findings`, `score`, `grade`, `badge_markdown`) instead of just a list of `Findings`. Default scans run 20 modules, excluding `cloud_recon`, `team`, and `nhi` for performance reasons. The `team` module is available but excluded from default scans as it manages local team membership rather than performing remote scanning.

Registry Functions

Function	Purpose	Returns
<code>build_module_registry()</code>	Builds remote module dict	<code>Dict[str, Dict]</code>
<code>build_local_modules()</code>	Builds local module dict	<code>Dict[str, Dict]</code>
<code>get_module_runner(id)</code>	Gets runner function by ID	Callable or None
<code>is_local_module(id)</code>	Checks if module is local	<code>bool</code>
<code>is_remote_module(id)</code>	Checks if module is remote	<code>bool</code>
<code>list_remote_modules()</code>	Lists all remote module IDs	<code>List[str]</code>
<code>list_local_modules()</code>	Lists all local module IDs	<code>List[str]</code>
<code>get_module_color(id)</code>	Gets display color for module	<code>str</code>

PART 4: Core Infrastructure Layer

__init__.py (73 LOC)

The package entry point defines version 11.0.0 and exports `scan`, `ReconProResult`, `audit_scan`, and `__version__` as the public API. The 67-line module docstring serves as the canonical catalog, listing all remote modules (13 listed, but 24 actual), advanced modules (12), local modules (3), powers (13), and intelligence systems (3) with descriptions and CLI usage examples. The banner text "Twenty-Three Blades" is stale, dating from an earlier version. Imports `scan` from `scanner.py` (synchronous), not `engine.py` (asynchronous).

constants.py (117 LOC)

The single source of truth for all shared constants. Defines `SEVERITY_LEVELS` (5 levels: critical through info), `GRADE_THRESHOLDS` (6 tiers: A+ through F), color mappings for Rich console output, SARIF level mappings, badge color mappings, default timeouts (8s), rate limits (10.0/s), body limits (16KB), and max workers (4). Path constants define `RECONPRO_HOME` (`~/reconpro`) with subdirectories for scans, plugins, memory, knowledge graph, and tools. The `USER_AGENT` string says "ReconPro/10.0" but `__version__` is "11.0.0", representing a stale version string that has not been updated.

utils.py (277 LOC)

Seventeen shared utility functions with zero side effects and no ReconPro-internal imports beyond constants. Key functions include `extract_host()` for URL parsing, `validate_target()` for input validation with private IP detection, `count_severities()` for duck-type finding counting, `sort_findings_by_severity()` for stable sorting, `severity_to_cvss()` and `severity_to_dread()` for framework mapping, `compute_score()` for additive point deduction from base 100, `compute_grade()` for numeric-to-letter grade conversion, and `badge_markdown()` for GitHub shields.io badge generation. Shannon entropy calculation, private IP detection, and URL joining utilities round out the collection. Notably, `severity_to_dread()` uses a lazy import from constants despite constants already being imported at the module level.

interfaces.py (91 LOC)

Defines 5 runtime-checkable Protocols (`ScanModule`, `FindingProcessor`, `ReportGenerator`, `EventEmitter`, `ConfigurationProvider`) and one ABC (`PluginInterface`). Type aliases are defined but loosely typed: `Findings = list` provides no type safety. The dual typing approach allows modules to use either structural typing (Protocols) or nominal typing (ABCs). Backward compatibility is maintained as existing modules do not need to implement any interface explicitly.

context.py (75 LOC)

Defines a `ScanContext` dataclass intended to replace the 4-argument pattern (`target`, `base_url`, `timeout`, `verify_tls`) with a single context object. Includes auto-populated fields for `trace_id` (12-char UUID hex), `started_at` timestamp, and lazily extracted `host/scheme/port`. The `to_args()` method provides backward compatibility. However, this context class is defined but NOT actually used by `scanner.py` or `engine.py`, representing a planned migration that was never completed. The `port` field is declared but never populated, always remaining 0.

delta.py (708 LOC)

Dynamic delta reporting system comparing two scan results with git integration. Provides `DeltaReporter` class with `compare()`, `generate_markdown()`, and `generate_sarif()` methods. Supports snapshot save/load/compare

operations persisted to `~/reconpro/snapshots/`. Integrates with git for blame attribution on new findings. Notable issues include duplicated `_SEV_ORDER` from `constants.py`, stale SARIF version "8.5.0" (should be 11.0.0), finding deduplication using title as key (fragile), and dead code in `_git_blame()` where summary parsing attempts to extract a hash from the summary field instead of the first output line.

formats.py (502 LOC)

Multi-format export system supporting SARIF 2.1.0, Markdown, JSON, HTML, and PDF (via print-ready HTML with `@media print` CSS). The `export_sarif()` function builds unique rules per category, maps findings to SARIF results with locations, properties, and evidence, and includes CVE data, intelligence scores, and attack path information. The `export_markdown()` function generates GitHub-style reports with badges, severity tables, module breakdowns, remediation sections, and intelligence analysis. Notable duplication: `_SARIF_LEVEL` and `_SEV_ORDER` are copied from `constants.py` instead of being imported.

PART 5: Dual Scan Engines

scanner.py - Synchronous Engine (210 LOC)

The original sequential scan orchestrator. Runs modules one-by-one in a for loop, collecting findings into ReconProResult. Creates a RateLimiter instance but never passes it to any module runner (dead code). Does NOT run the IntelligencePipeline post-scan, meaning results have intelligence=None when using this engine. Does NOT call HookManager for plugin integration. This is the engine imported by __init__.py and used by the CLI, making it the default execution path for reconpro.scan(). Special-cases the vibesec module which returns a 4-tuple instead of a list.

engine.py - Asynchronous Engine (727 LOC)

The concurrent scan orchestrator designed as a drop-in replacement for scanner.scan(). Uses asyncio.Semaphore for bounded concurrency (default 5). Implements ScanEvent-driven architecture with EventCollector for real-time progress tracking. Creates per-scan RateLimiter instances (fixed shared-state bug from prior sessions). Runs IntelligencePipeline.analyze() post-scan. Integrates HookManager for pre_scan, post_finding, and post_scan plugin lifecycle hooks. Module exceptions are caught and recorded as info-severity findings, meaning a crashing module silently downgrades to non-affecting results. The concurrent_scan() function has dead code: creates a global_semaphore that is never used, as each scan_one() creates its own ScanEngine with internal semaphore.

Feature	scanner.py (Sync)	engine.py (Async)
Execution	Sequential for-loop	Concurrent asyncio tasks
Intelligence Pipeline	No	Yes (post-scan)
Hook System	No	Yes (HookManager)
Rate Limiter	Created but unused	Per-scan instances
Event System	None	ScanEvent + EventCollector
Module Error Handling	Raises exception	Creates info-severity finding
CLI Default	Yes	No
vibesec Handling	4-tuple detection	4-tuple + fallback

[CRITICAL] Dual Engine Divergence: The two engines have fundamentally different behavior regarding intelligence analysis, plugin hooks, error handling, and rate limiting. This is a CRITICAL architectural issue identified in the Team 1 Architecture Review.

PART 6: CLI and TUI Systems

cli.py - Command Line Interface (2,014 LOC)

A massive 2,014-line file implementing 50+ subcommands via argparse. The main() function alone spans 1,765 lines, constituting a God function that should be decomposed into separate command modules. Uses Rich for terminal rendering with progress bars, tables, panels, and spinners. Imports MODULE_REGISTRY and other registries from scanner.py (which re-exports from registry.py), creating an indirect dependency chain. SEV_COLORS is hardcoded locally instead of imported from constants.py, creating a maintenance risk. Many advanced module commands follow identical patterns (banner -> lazy import -> spinner wrap -> print findings) that should be abstracted into a generic dispatcher. The implicit scan feature auto-parses positional arguments as scan targets when no subcommand is given.

Key CLI Commands

Category	Commands	Count
Scanning	scan, vibesec, audit, dev, doctor, blitz, ports, secrets	8
Advanced Modules	quantum-fingerprint, dark-web, info-ops, steg, covert, zero-day, ghost, sigint, attributor, weaponized-report, honeypot, dead-drop	14
Intelligence	agent, graph, nexus, ai-redteam, supply-chain, cross-validate	6
Recon & Intel	subdomains, passive, cve, threat-feeds, geoip	5
Infrastructure	serve, schedule, swarm, adversarial, deploy	5
Analysis	report, history, diff, delta, benchmark, rate, profile	7
Meta	list, open, plugin, chat, tui, export, zai, wishes, info	9

cli_help.py - Help Database (1,004 LOC)

A pure data file containing help text for 36+ modules and 25+ CLI commands, totaling approximately 700 lines of structured help data. Each module entry includes description, examples, details, output format, and cross-references. The render_module_help() function formats help boxes with DESCRIPTION, DETAILS, EXAMPLES, OUTPUT, and SEE ALSO sections. render_all_modules() produces a complete module listing, and render_quick_start() generates a 12-step getting-started guide.

nexus_tui.py - Terminal UI (3,359 LOC)

The second-largest file in the codebase implements a full Textual-based TUI application with 124 methods in the NexusApp class. Features include animated boot sequence, real-time module status indicators, pulsing status dots, scan progress tracking, session persistence to JSON, command completion with fuzzy matching, vim-style keybindings, and a responsive layout system. Seven custom widgets (ScoreGauge, Sparkline, StatCounter, VelocityMeter, CommandCompleter, HintBar, Toast) provide rich visual feedback. The TUI supports 20+ scan types including scan, audit, dev, doctor, blitz, subdomains, agent, swarm, adversarial, graph, passive, fuzzer, cve, profile, netmap, defense, compliance, delta, benchmark, iac, container, ast, cloud_recon.

chat.py - Interactive Chat Mode (448 LOC)

Natural language interface allowing operators to "talk to ReconPro like a teammate." Supports commands like scan, audit, dev, doctor, blitz, subdomains, agent, history, compare, report, score, secrets, ports, and screenshot. Bare domain/URL input is automatically treated as a scan target. Features ASCII art banner and

Rich-formatted colored output.

PART 7: Scanning Modules Inventory (28 Modules)

The scanning modules directory contains 30 Python files (28 modules + `__init__.py` + `container_sec.py`), totaling approximately 34,902 lines of code. All modules are fully implemented with complete logic and zero stubs. Every module exposes a `run_*(target, base_url, timeout, verify_tls) -> List[Finding]` entry point and uses the universal Finding dataclass from `http_layer.py` for output standardization. The average module is approximately 1,163 lines of code.

Core Remote Modules

Module	LOC	Category	Key Capabilities	Status
recon.py	1,247	Surface Recon	25-category recon: DNS, headers, tech, TLS, ports, WAFs	Implemented
auth.py	1,038	Authentication	JWT analysis, bypass, OAuth, session, rate limiting	Implemented
chain.py	291	SSRF/Redirect	Redirect chains, SSRF injection, open redirect	Implemented
bot.py	355	C2 Detection	Malware signatures, C2 paths, threat feed reputation	Implemented
gorgon.py	634	Adversarial	17-stage fear engine, DREAD scoring	Implemented
oblivion.py	748	Deep Oracle	23-stage DREAD analysis, 6 fear levels	Implemented
vibsec.py	206	Vibe Benchmark	7-category AI/vibe-coding vuln scoring	Implemented
nhi.py	159	Non-Human ID	Cloud metadata, token scanning, credential probing	Implemented
pegasus.py	195	Spyware	116 C2 domains, SMS patterns, backup analysis	Implemented
cloud_recon.py	1,152	Cloud Infra	AWS/Azure/GCP/DO metadata, S3, DNS	Implemented
team.py	256	Team Mgmt	Local team CRUD, activity logging, invite codes	Implemented

Advanced Remote Modules (v9.2.0)

Module	LOC	Key Innovation	Status
quantum_fingerprint.py	2,025	7-signal HTTP timing OS fingerprinting (novel)	Implemented
dark_web_monitor.py	788	7 paste sources, 6 OSINT APIs, breach DB	Implemented
free_info_ops.py	2,080	Deception resilience, false flag analysis	Implemented
steganography_detector.py	1,381	10 detection categories, chi-squared test	Implemented
covert_channel.py	1,673	8 channel classes, simulation + detection	Implemented
zero_day_hunter.py	1,576	Statistical anomaly detection, fuzzing analysis	Implemented
infrastructure_ghost.py	2,116	IP discovery, CT mining, drift detection	Implemented
signal_intelligence.py	2,046	9 C2 frameworks, beaconing, traffic analysis	Implemented
nation_state_attributor.py	769	22 APT groups, 5 attribution methods	Implemented
weaponized_report.py	2,509	50+ tracker signatures, stego watermarks	Implemented
honeypot_dance.py	1,815	Response timing, 15+ honeypot fingerprints	Implemented
dead_drop.py	2,222	8 crypto channels, DNS/TXT/ETag/CT detection	Implemented

Local Audit Modules

Module	LOC	Platform	Checks	Status
host.py	1,488	Linux/macOS/Win	27: ports, SSH, Docker, env, SUID, kernel CVEs	Implemented
dev.py	1,087	All	19: deps, git, .env, secrets, Docker, lockfile CVEs	Implemented
doctor.py	1,256	Linux/macOS/Win	17: password, encryption, antivirus, firewall	Implemented

Special Modules

iac_audit.py (2,121 LOC) parses Terraform, CloudFormation, Dockerfiles, K8s manifests, and Docker Compose files with custom parsers that require no external YAML library. container_sec.py (1,018 LOC) analyzes Dockerfiles and K8s manifests for container escape vectors with compound risk evaluation. ast_analyzer.py (611 LOC) performs Python AST walking with 20 security checks plus JS/TS regex scanning with 14 patterns.

PART 8: Intelligence Systems (3 Engines)

The intelligence layer consists of three engines that compose into a unified Intelligence Pipeline. All three are rule-based (no LLM required) and use only Python standard library. They are orchestrated by `intelligence_pipeline.py` which runs them sequentially and computes composite risk scores.

AI Analyst Engine (`ai_analyst.py` - 1,639 LOC)

A pure-Python rule-based security analysis engine implementing 6 sub-engines and 12 classes. The `FindingClassifier` uses 35 regex patterns and 17 module rules with 4-phase scoring (title match, module source, category hint, evidence enrichment). The `FindingCorrelator` uses SHA256 deduplication and $O(n^2)$ pair analysis capped at 50,000 pairs to detect shared-asset correlations and attack chains across 17 chain templates. The `ExploitabilityEstimator` provides category-specific baseline scores for 14 vulnerability types. The `BusinessImpactAnalyzer` assesses 5 impact dimensions (financial, reputational, operational, compliance, legal). The `AttackPathDetector` uses DFS/BFS graph traversal across 11 multi-step chain templates. The `RemediationPrioritizer` ranks fixes by weighted priority (severity $\times 0.35$ + exploitability $\times 0.25$ + impact $\times 0.25$ + confidence $\times 0.15$). Known issue: `_detect_chains` line 662 has "if shared or True:" which always executes regardless of the shared check (debug leftover).

Attack Graph Engine (`attack_graph.py` - 988 LOC)

Builds directed attack graphs from scan findings using a custom `DiGraph` implementation with adjacency lists. Supports BFS path finding, DFS all-paths enumeration, approximate betweenness centrality, choke point detection (30% path concentration threshold), and single-point-of-failure identification via path-block simulation. Maps findings to 7 Lockheed Martin kill chain phases via 30 category-to-phase mappings. Uses 33 relationship templates defining how vulnerability categories connect. Maximum graph size: 5,000 nodes and 20,000 edges with safe edge addition. This is the CRITICAL BOTTLENECK scaling at $O(n^2)$, consuming 87.6% of pipeline time at 2000 findings.

Threat Intel Engine (`threat_intel.py` - 940 LOC)

Centralized threat intelligence enrichment engine with thread-safe caching and TTL eviction. Contains embedded databases: 25 known CVE entries (Log4Shell, XZ Utils Backdoor, etc.), CISA KEV list, CWE mappings for 24 categories, MITRE ATT&CK technique mappings, CAPEC mappings, and OWASP Top 10 with full descriptions. Implements 32 technology detection regex patterns and online NVD API lookup with SSRF protection (host validation, HTTPS-only, 65KB response cap). Scales linearly $O(n)$ making it the most efficient of the three engines.

Intelligence Pipeline (`intelligence_pipeline.py` - 456 LOC)

Post-scan orchestration composing all three engines. Supports selective engine activation via flags. Computes 5 composite scores: `executive_risk_score` (severity $\times$ count_factor + chain_boost), `exposure_score` (critical $\times 15$ + high $\times 8 \times$ asset_factor), `mission_impact_score`, `infrastructure_health_score` (100 - executive_risk), and `threat_confidence_index`. Risk levels: CRITICAL (>75), HIGH (>55), MEDIUM (>35), LOW (>15), MINIMAL. Maximum finding cap: 10,000. All engine failures are caught and reported as errors without crashing the pipeline.

Intelligence Pipeline Performance Benchmarks

Finding Count	AI Analyst (ms)	Attack Graph (ms)	Threat Intel (ms)	Pipeline (ms)
100	205	84	36	312

Finding Count	AI Analyst (ms)	Attack Graph (ms)	Threat Intel (ms)	Pipeline (ms)
500	1,089	1,835	168	3,188
1,000	1,966	7,493	342	9,977
2,000	3,772	30,973	709	36,968

PART 9: Security Architecture

security.py (811 LOC)

Centralized security hardening layer providing multi-layered input validation, safe parsing, secret detection, and audit logging. Implements 5 sanitization functions (target, path, filename, HTML, shell), 3 safe parsers (JSON with 1MB/20-depth/10K-key limits, URL with scheme whitelist, XML with entity expansion prevention), 10 regex-based secret detection patterns, structured JSON audit logging with rotating file handler (10MB, 5 backups), file hash computation, and supply chain integrity checking. Classifies threats into 5 categories: injection, crypto, network, data, and auth.

defense.py (1,402 LOC)

Autonomous defense generation producing deployable WAF rules (ModSecurity, Nginx, Cloudflare), code patches with before/after diffs, Terraform infrastructure fixes, and remediation text for 20+ vulnerability categories. Uses template-based generation with variable substitution. DefenseBundle computes a defense readiness score 0-100.

sanitize.py (32 LOC)

Convenience re-export facade providing clean import path from reconpro.sanitize to 11 security.py functions. No original code. Pure re-export module.

entropy.py (727 LOC)

Granular secret entropy scoring combining Shannon entropy with contextual signals. Contains 50+ known secret prefixes (AWS, GitHub, GitLab, Slack, Stripe, Twilio, Google, JWT, Azure) with format validation for 12+ prefix types. 8-rule classification matrix produces severity levels (CRITICAL through INFO). Supports file and directory scanning with 29 default extensions, skipping 20+ blacklisted directories and binary files.

Security Audit Findings (Team 3)

Severity	Count	Key Issues
CRITICAL	3	Path traversal in /report and /history, unauthenticated token generation
HIGH	6	XSS in HTML reports, unsandboxed plugins, SSRF, shell=True, error leakage
MEDIUM	10	CORS wildcard, unbounded SSE/token dict, no rate limiting on scan endpoint
LOW	6	Sensitive MITM captures, missing audit validation, stale code

PART 10: Network and Protocol Layer

http_layer.py (165 LOC)

The universal HTTP abstraction used by every scanning module. Defines the Finding dataclass (10 fields: title, severity, category, module, description, evidence, asset, points_deducted, remediation, dread_score), a token-bucket RateLimiter, and the http_probe() function with hardened error handling, 16KB body limit, and TLS configuration. Also provides compute_grade() and badge_markdown() utility functions. All modules import from this file, making it the most depended-upon file in the codebase.

Module	LOC	Purpose	Key Feature
async_http.py	902	Async HTTP with adaptive rate limiting	Per-domain EMA, aiohttp->urllib fallback
connection_pool.py	255	Thread-safe HTTP pool with SSL caching	Double-checked locking, perf counters
proxy.py	705	Proxy pool with rotation strategies	4 strategies, Tor circuit management
evasion.py	1,117	Anti-fingerprinting and WAF bypass	40+ UAs, JA3/JA4, timing jitter
raw_sockets.py	598	SYN scan, UDP probe, OS fingerprint	Scapy fallback, 3-method TTL analysis
tunnel_detect.py	1,025	Protocol tunnel detection	8 tunnel types, entropy analysis
mitm.py	612	MITM proxy and traffic analysis	15 token patterns, auth bypass testing
browser_mod.py	158	Playwright browser automation	JS-rendered secret scanning

Three competing HTTP layers exist: http_layer.py (synchronous, used by all modules), async_http.py (async with adaptive rate limiting), and connection_pool.py (thread-safe pool with SSL context caching). Each has different User-Agent strings: ReconPro/10.0, ReconPro/10.0, and ReconPro/2.0 respectively. The async_http.py module provides the most sophisticated implementation with per-domain adaptive rate limiting using exponential moving average for response time and error rate, automatic back-off on 429/503 responses with jitter, concurrent limit reduction on timeouts, and manual redirect following with 303->GET conversion.

PART 11: Report Generation System

report_writer.py (551 LOC)

AI-powered report generation with optional LLM enhancement (OpenAI) and template fallback. Supports 4 audience modes: executive (max 5 findings, business risk focus), technical (all findings), compliance (framework-mapped with ISO 27001/NIST 800-53), and developer (actionable PRs). The ExecutiveSummaryGenerator tries LLM first, falls back to templates. Template methods generate structured markdown with severity overviews, module breakdowns, and priority remediation lists.

reports.py (1,959 LOC)

Chart.js-powered HTML report generation with interactive visualizations. Generates 8 chart types: score gauge, severity doughnut, DREAD radar, module bar, points horizontal bar, category bar, severity stacked bar, and DREAD grouped bar. All charts use CDN-loaded Chart.js 4.4.7. The HTML report is a single self-contained file with inline CSS, glassmorphism cards, gradient accent lines, animated stat counters, smooth scroll navigation, and responsive design. Also generates CSV, SARIF, XML, Markdown, and JSON reports. The PDF generation function is a placeholder that generates print-ready HTML instead of actual PDF.

Report Formats Summary

Format	Engine	Features	Status
SARIF 2.1.0	formats.py	Full rules, locations, CVE data, intelligence	Complete
Markdown	formats.py / report_writer.py	GitHub badges, tables, remediation	Complete
HTML Interactive	reports.py	Chart.js, glassmorphism, animated	Complete
JSON	formats.py	Pretty-printed, all fields	Complete
PDF (print HTML)	formats.py	@media print CSS, styled	Placeholder
CSV	reports.py	Tabular data export	Complete
XML	reports.py	Structured data export	Complete

PART 12: Integration Layer (6 Clients)

The integration layer provides 6 enterprise clients, all using only Python stdlib (urllib.request for HTTP, json for serialization). All clients implement a consistent `sync_findings(findings, target)` interface returning structured dicts. Configuration is file-based (YAML or JSON) stored at `~/.reconpro/integrations/`.

Client	LOC	Auth	Config	Key Features
JiraClient	391	Basic (email:token)	YAML	ADF descriptions, issue sync, JQL search
SlackClient	411	Webhook/Bot Token	YAML	Block Kit alerts, daily digest, mrkdwn
GitHubClient	508	Bearer token	YAML	SARIF upload, PR comments, commit status
SplunkClient	54	HEC token	Explicit	Minimal HEC sender
PagerDutyClient	72	API key	Explicit	Incident creation (5 cap)
ZAISStreamClient	428	Bearer + SSE	Auto-discover	Streaming analysis, chat, health check

Notable patterns: Jira, Slack, and GitHub clients share identical `_load_config()` functions (copy-pasted across 3 files, should be extracted). The ZAISStreamClient is the most complex integration with SSE parser, streaming support, and auto-config discovery from three search paths. Splunk and PagerDuty are minimal implementations without file-based config loading. Error handling is inconsistent: Jira/GitHub/Slack raise `RuntimeError` while Splunk/PagerDuty return error dicts.

PART 13: Knowledge Graph and Social Graph

knowledge_graph.py (774 LOC)

Directed security knowledge graph supporting 11 node types (target, subdomain, port, service, vulnerability, finding, CVE, asset, endpoint, technology, certificate) and 9 edge types. Features BFS-based attack surface analysis (depth-3), blast radius computation (depth-10), DFS attack chain discovery (max 8 hops), cross-correlation via shared technology/port, and Neo4j-compatible Cypher output. NetworkX is optional with a complete pure-Python fallback implementation (_FallbackDiGraph). The dual-backend approach results in approximately 20 occurrences of if/else branching that could be abstracted behind a unified graph interface. Persistence via JSON serialization with NetworkX node-link format or fallback dict format.

social_graph.py (935 LOC)

OSINT social graph intelligence extracting entities and relationships from 4 passive data sources: DNS resolution, WHOIS records, HTTP headers, and crt.sh certificate transparency logs. Detects CDN providers (11: Cloudflare, AWS CloudFront, Fastly, Vercel, etc.), proxy chains, and co-hosted domains. Implements Tarjan's articulation point algorithm ($O(V+E)$) for bridge entity detection, unweighted BFS shortest path, and connected component clustering. WHOIS extraction depends on the system whois command which may not be available on all platforms. crt.sh queries are not rate-limited.

graph_ui.py (678 LOC)

Generates self-contained HTML files with D3.js force-directed graph visualization. All JavaScript and CSS embedded inline (no external dependencies). Node coloring by type (target=blue, vulnerability=red, CVE=orange, technology=gray, etc.). Interactive features: hover tooltips, zoom, pan, click details.

PART 14: Kill Chain Engine

The `kill_chain.py` module (2,992 LOC) implements a full 7-phase Lockheed Martin Cyber Kill Chain automation with the largest single class in the codebase (`KillChainEngine` with 54 methods). Each phase maps to MITRE ATT&CK; techniques: Reconnaissance (5 techniques), Weaponization (4), Delivery (6), Exploitation (6), Installation (5), Command & Control (5), and Actions on Objectives (4).

Phase	Methods	Key Techniques
Reconnaissance	6	DNS harvest, WHOIS, subnet enum, subdomain extract, ASN lookup, tech detect
Weaponization	3	SQLi/XSS/LFI payload generation
Delivery	7	MX/SPF/DKIM, upload endpoints, phishing, SSRF, open relay
Exploitation	10	Remote services, public apps, sensitive paths, backups, DB dumps, cron
Installation	7	Persistence, C2 paths, botnet, reverse shell, WebSocket, proxy, encrypted
Command & Control	3	Exfiltration risk, lateral movement, priv esc paths
Actions on Objectives	4	Business impact, data classification, DoS, alternate exfil

The `ThreatMatrix` cross-references phases with MITRE ATT&CK; tactics, computing weighted overall risk (exploit=0.25, install=0.20, delivery=0.15, C2=0.15, weaponize=0.10, actions=0.10, recon=0.05). The `AttackTree` builds an OR-tree representation from phase data. Technology signatures for 8 platforms (WordPress, Drupal, Joomla, Apache, Nginx, IIS, PHP, ASP.NET) include actual exploit payloads for testing purposes. The `C2_PATH_SIGNATURES` constant contains 21 common C2 URI patterns for detection.

PART 15: Advanced Capabilities

Module	LOC	Purpose	Notable
attribution.py	3,145	Nation-state attack attribution	22 APT groups, 14 campaigns, 5 signals
fuzzer.py	1,744	Context-aware payload fuzzing	Tech stack detection, payload selection
chain_engine.py	2,228	Multi-vector attack path chaining	50+ rules, graph-based, LLM-assisted
adversarial.py	1,150	3-agent remediation loop	HACKER/CODER/GUARDIAN, 20+ rules
swarm.py	1,121	Multi-agent multiprocessing	4 agents (SCOUT/HACKER/CODER/GUARDIAN)
webhooks.py	876	VCS webhook listeners	HMAC-SHA256, PR comments, commit status
api_discovery.py	1,185	Dynamic API blueprint reconstruction	6 sub-engines, GraphQL, gRPC, OpenAPI
netmap.py	724	Zero-trust network mapping	ICMP/ARP/mDNS/port scan, trust mapping
geoip.py	684	IP geolocation enrichment	Batch API, cloud provider heuristics, cache
shadow_it.py	1,684	Shadow IT discovery	Abandoned infrastructure, orphaned DNS, credential leaks
memory.py	903	Unified memory store	FindingStore, AgentBlackboard, CredentialVault
exfil_channels.py	974	Exfiltration channel mapping	Channel viability scoring, risk assessment
wishes.py	Large	22-wish ritual orchestration	Fear Index, witness records, hall of fame
nexus_agent.py	3,417	Agentic security engine	22 tools, LLM/rule-based, CVSS scoring

The attribution.py module is the third-largest file at 3,145 LOC, implementing TTP matching with Jaccard similarity, campaign tracking, nation-state profiling via weighted voting (ASN 0.25, TLS 0.25, language 0.20, WAF 0.20, timezone 0.10), and country-specific signatures for CN, RU, IR, KP, VN, PK. The nexus_agent.py is the largest file at 3,417 LOC, serving as the core intelligence brain with 22 registered tools, CVSS v3.1 scoring (16 category profiles), 25 correlation rules, and support for both OpenAI and Anthropic LLM backends with pure rule-based fallback.

PART 16: Observability and Telemetry

The observability layer implements structured logging, metrics collection, distributed tracing, and health monitoring using only Python stdlib. No external observability frameworks (OpenTelemetry, Prometheus client, etc.) are used.

Component	Purpose	Key Features
StructuredLogger	JSON logging with trace context	File output, rotating handler, exception logging
MetricsCollector	Counters, gauges, histograms, timers	Thread-safe, snapshot, reset, disabled mode
ScanTracer	Distributed tracing for scans	Span timing, module tracking, trace propagation
PerformanceProfiler	Code profiling	Function timing, call counting, memory tracking
HealthMonitor	Component health checking	Periodic checks, degradation detection
TelemetryManager	Unified telemetry API	Convenience methods, global singleton

Testing coverage for observability is excellent with 132 test methods covering all components, including thread safety verification, disabled mode behavior, file output validation, exception handling, and convenience API surface.

PART 17: Testing Infrastructure

The test suite consists of 29 test files with approximately 1,250+ test methods totaling 11,185 lines of code. All tests use unittest.TestCase (stdlib). No pytest framework or external mocking libraries are used (except unittest.mock in diagnostics and performance tests). Every test file manually inserts the project root into sys.path.

Test File	Tests	Coverage Area	Assessment
test_security.py	181	All security.py functions	Very Comprehensive
test_observability.py	132	All observability components	Very Comprehensive
test_utils.py	118	All 17 utility functions	Very Comprehensive
test_coverage_boost.py	110	Edge cases, boundary values	Comprehensive
test_diagnostics.py	103	Health checks, version info	Comprehensive
test_ai_analyst.py	81	All AI analyst sub-engines	Comprehensive
test_constants.py	44	Every constant group	Comprehensive
test_threat_intel.py	48	Databases, enrichment, engine	Comprehensive
test_interfaces.py	79	All protocols, ABCs, context	Comprehensive
test_cli.py	47	Rendering, commands, edge cases	Comprehensive
test_widgets/*	0	TUI widgets	NO COVERAGE
test_integrations/*	0	Jira, Slack, GitHub clients	NO COVERAGE

[CRITICAL] Zero test coverage for the entire TUI widget layer (7 widgets, ~2,000 LOC) and all 6 enterprise integration clients (~1,900 LOC).

PART 18: Deployment and Configuration

Docker Configuration

Production-grade multi-stage Dockerfile using python:3.12-slim builder. Runtime uses non-root user (UID/GID 1001), PYTHONUNBUFFERED=1, and installs reconpro[full] with rich, textual, requests, and optional aiohttp. Health check via curl to localhost:7890 every 30s. Entrypoint: reconpro serve --port 7890. docker-compose.yml provides resource limits (2 CPU / 1GB RAM), security hardening (no-new-privileges, cap_drop ALL), custom bridge networking (172.28.0.0/16), named volume, JSON logging (50MB, 5 files), and horizontal scaling support.

Kubernetes Manifests

Production-grade K8s deployment with 3 replicas, RollingUpdate strategy, security context (runAsNonRoot, seccomp RuntimeDefault, no privilege escalation), liveness/readiness/startup probes, resource requests (250m/256Mi) and limits (1/512Mi), PVC for persistent data, topology spread constraints, pod anti-affinity, tolerations for dedicated nodes, and Prometheus scrape annotation. Service is ClusterIP with topology-aware routing hints. Namespace: reconpro.

pyproject.toml

Declares version 10.0.0 (contradicts deploy configs at v11). Requires Python >=3.8. Core dependencies: rich>=13.0.0, textual>=0.40.0, requests>=2.28.0 (contradicts ADR-001 zero-dependency claim). Optional groups: async (aiohttp), browser (playwright), llm (openai, anthropic), graph (networkx), raw (scapy), intel (shodan), collab (websockets), integrations (jira, slack-sdk), and full (all combined). Entry point: reconpro = reconpro.cli:main.

[CRITICAL] Version Inconsistency: pyproject.toml says 10.0.0, code says 11.0.0, deploy says v11, README says v9.0.0

PART 19: Known Issues and Technical Debt

Critical Issues

ID	Severity	Issue	Location	Status
C-01	CRITICAL	Path traversal in /report endpoint	server.py	Fixed
C-02	CRITICAL	Path traversal in /history endpoint	server.py	Fixed
C-03	CRITICAL	Unauthenticated token generation	server.py	Known
C-04	CRITICAL	Dual engine divergence	scanner.py/engine.py	Known
C-05	CRITICAL	Version inconsistency (4+ versions)	7+ files	Known

Code Duplication (~274 LOC)

Duplication	Files	Lines	Recommendation
SEVERITY_LEVELS/_SEV_ORDER	constants, delta, formats	~15	Import from constants
SARIF_LEVEL_MAP	constants, delta, formats	~12	Import from constants
SEV_COLORS/GRADE_COLORS	constants, cli, reports	~20	Import from constants
compute_grade()	utils, http_layer, formats	~30	Import from utils
_load_config()	jira, slack, github	~45	Extract shared utility
Finding classification	ai_analyst, attack_graph, threat_intel	~100	Extract shared classifier
User-Agent strings	http_layer, connection_pool, async_http, proxy	~10	Centralize in constants

Dead Code and Unused Features

- ScanContext in context.py - defined but never used by any scanner
- global_semaphore in engine.py concurrent_scan() - created but never used
- RateLimiter in scanner.py line 86 - instantiated but never passed to modules
- Dead code in delta.py _git_blame() lines 125-126
- ai_analyst.py line 662: "if shared or True:" debug leftover
- port field in context.py - declared but always 0
- generate_pdf_report in reports.py - placeholder returning HTML
- README.md - stale v9.0.0 placeholder with no content

PART 20: Roadmap and Recommendations

Prioritized Recommendations

Priority	Recommendation	Impact	Effort
P0	Unify version strings across all files to single source of truth	Critical	Low
P0	Consolidate dual scan engines into single engine with sync/async modes	Critical	High
P0	Eliminate constant duplication (import from constants.py everywhere)	High	Low
P1	Fix unauthenticated token generation (C-03 security issue)	Critical	Low
P1	Add test coverage for TUI widgets (0% currently)	Medium	Medium
P1	Add test coverage for integration clients (0% currently)	Medium	Medium
P1	Optimize Attack Graph $O(n^2)$ to asset-indexed adjacency	High	High
P2	Extract shared _load_config from integration clients	Low	Low
P2	Complete ScanContext migration in scanner/engine	Medium	Medium
P2	Implement real PDF generation (replace HTML placeholder)	Low	Medium
P2	Decompose cli.py main() (1,765 lines) into command modules	Medium	Medium
P3	Abstract graph backend (NetworkX vs fallback) into interface	Low	Medium
P3	Replace custom YAML parser with PyYAML dependency	Low	Low
P3	Add finding UUID for proper deduplication	Medium	Medium

Roadmap Alignment

The existing ROADMAP.md describes v10 as current and v11 as future, but the code already identifies as v11 in constants.py and deploy manifests. Key planned features for v11 include: distributed agent mesh, real-time collaborative scanning, plugin marketplace with signing, GraphQL API, web dashboard, cross-scan correlation engine, active exploitation, network-level scanning (raw sockets), wireless assessment, cloud API testing, RBAC, compliance mapping (SOC2/ISO 27001/PCI DSS), and multi-tenant MSSP support. Many of these capabilities already exist in partial form (collab.py for collaboration, graph_ui.py for dashboard, raw_sockets.py for network scanning), suggesting the roadmap should be updated to reflect current implementation status.

Competitive Positioning

ReconPro occupies a unique niche with no direct open-source equivalent. The combination of 28 modules, zero external dependencies, pure-Python portability, and integrated intelligence pipeline (AI analyst + attack graph + threat intel) is unmatched. Key differentiators with no OSS equivalent include: HTTP timing OS fingerprinting (novel approach), nation-state attribution (22+ APT groups), covert channel detection (8 classes), infrastructure ghosting, integrated DREAD scoring, and the 22-wish ritual orchestration system. The primary competitive weakness is the lack of persistent data storage and the absence of STIX/TAXII output format for threat intelligence sharing.